

Guided-Wave Structural Health Monitoring of Metallic Plates: Method Review, Finite-Element Crack Detection, and Four-Transducer Crack Localisation

Rouzbeh Pourjafarian

Department of Aerospace Engineering, Sharif University of Technology, Tehran, Iran

Abstract

This report documents a project on structural health monitoring (SHM) of thin metallic plates using piezoelectric transducers and ultrasonic guided waves, carried out in three phases. The project opens with a survey of the available monitoring methods, from vibration-based techniques through acoustic emission, fibre-optic sensing, electromechanical impedance and thermography, which ends in a justification for choosing the guided-wave route. It then builds a finite-element model of an aluminium plate carrying two bonded PZT discs, in which the piezoelectric patches are solved implicitly and the plate explicitly through an Abaqus co-execution. Comparing the sensor voltage from a healthy plate against a plate with a 25 mm crack gives a signal difference coefficient of 0.989 and a peak-amplitude drop of 19.5%, which establishes that the crack is detectable but says nothing about where it is. The project's final phase extends the model to a four-transducer network on a 400×400 mm plate, which yields six independent sensing paths. Damage indices built from the continuous wavelet energy of each path are fed to a probability-based diagnostic imaging (PDI) algorithm, and the crack is located to within 10 mm of its true position, about 2.5% of the plate span. A time-of-flight delay-and-sum image built from the same signals is also reported, together with a full-field wavefield-difference check that confirms the crack is represented correctly in the model itself.

Keywords: structural health monitoring, Lamb waves, piezoelectric transducers, finite element modelling, damage index, crack localisation, probability-based diagnostic imaging, Abaqus

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1. Introduction

Structures in service accumulate damage, and the traditional answer to that is periodic manual inspection. Scheduled inspection has two weaknesses. It only sees the structure at fixed intervals, so damage that initiates and grows between two inspections goes unnoticed, and it depends heavily on the judgement of the inspector. Structural health monitoring replaces that with permanently installed sensors and an automated diagnostic algorithm, so the condition of the structure can be interrogated whenever it is needed rather than when the maintenance calendar allows.

The aerospace case is the one that motivated this work. Airframes are thin-walled, fatigue-critical, and expensive to disassemble for inspection. A method that can interrogate a large area of skin from a small number of permanently bonded transducers avoids most of that disassembly, and ultrasonic guided waves are the main candidate for the role. In a plate whose thickness is comparable to the acoustic wavelength, the wave is guided by the two free surfaces and travels tens of centimetres with modest attenuation, which means a sparse network of transducers can cover a wide area. Any crack, delamination or corrosion patch along the path between two transducers scatters the wave and changes the received signal.

1.1. Scope and structure of this report

The work reported here was carried out in three stages, and the report follows the same order.

1. Section 2 surveys the families of SHM methods and compares them against the requirements of a thin metallic aerospace panel, which leads to the choice of piezoelectric guided-wave monitoring. Section 3 then sets out the physical basis: the piezoelectric constitutive equations and Lamb-wave dispersion in a plate.
2. Section 4 builds a pitch-catch finite-element model in Abaqus with one actuator and one sensor, runs it in healthy and cracked states, and compares the two received signals using time-domain damage indices. The question it answers is whether damage is present.
3. Section 5 rebuilds the model with four transducers, giving six independent sensing paths, and combines the per-path damage indices into a two-dimensional damage-probability map. The question it answers is where the damage is.

Sections 4 and 5 are reported as a single progression rather than as separate exercises. Detection is established first, because a method that cannot detect damage has nothing to localise, and the four-transducer model then extends the same finite-element approach with the transducers, excitation design and crack representation that localisation requires. This report discusses each of those additions where it arises.

2. Structural health monitoring: levels, methods and selection

2.1. Levels of diagnosis

SHM tasks are usually organised into a hierarchy of increasing diagnostic ambition. The classification introduced by Rytter defines four levels, and later work has added two more [1, 2].

Table 1: Diagnostic levels in structural health monitoring.

Level	Question answered	Typical output
1	Is there damage?	Binary or scalar flag; novelty detection against a threshold
2	Where is it?	Coordinates, region or a damage-probability map
3	How severe is it?	Crack length, stiffness loss, section-area reduction
4	What is the remaining life?	Growth prediction, time to an unacceptable state
5	What type of damage is it?	Crack, corrosion, delamination, loose joint
6	What should be done?	Maintenance decision, load restriction

The distinction matters for this report because the three stages of work map onto the first two levels. The two-transducer model of Section 4 reaches Level 1, and the four-transducer network of Section 5 reaches Level 2. Levels 3 and above were not attempted.

2.2. Survey of monitoring methods

Vibration-based methods. The structure is excited at low frequency and its modal properties, natural frequencies, mode shapes, modal curvature or flexibility, are tracked over time. Damage reduces local stiffness and shifts those properties [4, 5]. The instrumentation is cheap and the method covers

the whole structure at once, which is why it dominates civil-infrastructure monitoring. The weakness is sensitivity: global modal parameters respond weakly to small local damage, and they respond strongly to temperature, which makes early crack detection in a metallic panel difficult.

Acoustic emission.. Rather than sending a wave in, acoustic emission listens for the elastic energy released by the damage itself as it grows [6, 7]. It is passive and can catch damage at the moment of initiation, which no active method can do. In exchange it only sees damage while it is actively growing, it needs continuous listening at high sample rates, and it is vulnerable to operational noise.

Ultrasonic guided waves.. A transducer launches a Lamb wave into the plate, and other transducers record what arrives [8, 22]. Because the wave travels far in a thin-walled structure, a small number of transducers covers a large area, and because the excitation frequency is chosen by the operator the wavelength can be matched to the size of the damage that must be found. The costs are a dispersive, multi-modal wavefield that has to be understood before the signals can be interpreted, and a strong dependence on a healthy-state baseline that itself drifts with temperature.

Fibre-optic sensing.. Fibre Bragg gratings measure strain at discrete points along a single fibre, and Brillouin or Rayleigh scattering methods give strain distributed continuously along its length [10, 11, 9]. The fibres are light, immune to electromagnetic interference and can be embedded in a composite laminate. They measure strain, though, and strain is a local quantity: damage away from the fibre produces little signal.

Electromechanical impedance.. A bonded PZT patch is driven over a frequency sweep, typically above 30 kHz, and its electrical impedance is measured. Because the patch is coupled to the structure, the electrical impedance carries a copy of the local mechanical impedance, and damage near the patch changes it [12]. The method is very sensitive but only within a small radius of each patch, which makes area coverage expensive.

Vision-based and thermographic methods.. Camera-based inspection detects surface cracks and measures displacement fields, and infrared thermography detects subsurface defects through the thermal contrast they produce under active heating [13, 14]. Both need line of sight and controlled conditions, so they suit periodic inspection better than permanent monitoring.

Machine-learning approaches.. Machine learning is a processing layer applied on top of any of the sensing methods above, to classify damage, detect novelty, or fuse multiple data sources [17, 15]. It becomes attractive once a large body of labelled signals exists, which in practice usually means simulated signals.

2.3. Selection

Table 2 summarises the comparison against the requirements of a thin metallic panel: sensitivity to a small crack, area covered per sensor, and suitability for permanent installation.

Table 2: Comparison of the main SHM method families for thin metallic panels.

Method	Small-crack sensitivity	Area per sensor	Permanent install
Vibration-based	Low	Whole structure	Yes
Acoustic emission	High *	Large	Yes
Guided waves (PZT)	High	Large	Yes
Fibre optic (FBG)	Medium	Local	Yes
Electromechanical impedance	High	Very local	Yes
Thermography / vision	Medium	Surface only	No

* Only while the damage is actively growing.

Guided-wave monitoring with bonded piezoelectric transducers is the only entry that combines high sensitivity to a small crack with a large area per sensor and permanent installation. The same transducer can act as actuator and as sensor, and the hardware is thin, light and cheap enough to be distributed over a structure [18, 19]. That is the method used for the rest of this report.

3. Physical basis

3.1. Piezoelectric constitutive equations

A piezoelectric material couples its mechanical and electrical states. In the strain-charge form, the strain tensor $\mathbf{S}$ and the electric displacement $\mathbf{D}$ are

$$\mathbf{S} = \mathbf{s}^E \mathbf{T} + \mathbf{d}^T \mathbf{E} \quad (1)$$

$$\mathbf{D} = \mathbf{d} \mathbf{T} + \boldsymbol{\epsilon}^T \mathbf{E} \quad (2)$$

where $\mathbf{T}$ is stress, $\mathbf{E}$ the electric field, $\mathbf{s}^E$ the compliance at constant field, $\boldsymbol{\epsilon}^T$ the permittivity at constant stress, and $\mathbf{d}$ the piezoelectric strain-charge coupling matrix. The two equations are what make a single patch usable in both directions. Applying a voltage produces a strain, so the patch acts as an actuator; deforming the patch produces a charge, so the patch acts as a sensor.

For a disc poled through its thickness, the components that matter are d_{33} , which couples the poling direction to itself, $d_{31} = d_{32}$, which couples the in-plane directions to the poling direction and is what converts an in-plane Lamb wave into a measurable voltage, and the shear terms $d_{15} = d_{24}$. The values used throughout this work are those of APC850, listed in Table 5.

3.2. Lamb waves in a thin plate

In a plate bounded by two traction-free surfaces, the bulk longitudinal and shear waves reflect and interfere until only certain guided modes survive. These are the Lamb modes, and they separate into symmetric and antisymmetric families governed by the Rayleigh–Lamb equations,

$$\text{symmetric:} \quad \frac{\tan(qh)}{\tan(ph)} = -\frac{4k^2pq}{(q^2 - k^2)^2} \quad (3)$$

$$\text{antisymmetric:} \quad \frac{\tan(qh)}{\tan(ph)} = -\frac{(q^2 - k^2)^2}{4k^2pq} \quad (4)$$

with

$$p^2 = \left(\frac{\omega}{c_L}\right)^2 - k^2, \quad q^2 = \left(\frac{\omega}{c_T}\right)^2 - k^2, \quad (5)$$

where h is the half-thickness, k the wavenumber, ω the angular frequency, and c_L , c_T the bulk longitudinal and shear velocities. These are transcendental in k and have to be solved numerically at each frequency. Doing so gives the phase velocity $c_p = \omega/k$ and the group velocity $c_g = d\omega/dk$ for each mode.

Two consequences shape the modelling decisions that follow. First, the velocities depend on frequency, so a pulse of finite bandwidth spreads as it travels and the received packet is wider than the transmitted one. Second, at least two modes exist at any frequency, the fundamental symmetric S_0 and antisymmetric A_0 , and they travel at different speeds, so the sensor sees them

as separate arrivals. Keeping the excitation below the cut-on frequency of the higher modes keeps the picture to those two.

For the 1.6 mm aluminium plate used in this work, driven at 200 kHz, solving the equations above gives the values in Table 3.

Table 3: Lamb-wave properties of the 1.6 mm aluminium plate at 200 kHz.

Mode	Phase velocity (m/s)	Group velocity (m/s)	Wavelength (mm)
S_0	5348	5332	26.7
A_0	1580	2627	7.90

The S_0 mode is the faster of the two and arrives first, which is why every damage index in this report is computed on the first arriving packet. The A_0 mode has the shorter wavelength, 7.90 mm against 26.7 mm, so it is the mode that sets the element size the mesh must resolve.

4. Stage 1: crack detection with a two-transducer path

If a wave is launched from one bonded transducer and received by another, does a crack on the path change the received signal enough to be measured? That is the only question the first model was built to answer.

4.1. Model description

An aluminium plate carries two PZT discs, one acting as actuator and one as sensor, placed on opposite faces at either end of the plate so that the direct path between them passes over the centre. The crack sits at that centre, directly across the path. Geometry is given in Table 4 and material data in Table 5.

Table 4: Geometry of the two-transducer model.

Aluminium plate	
Dimensions	$177.8 \times 76.2 \times 1.6$ mm
Centre	$(0, 0)$ mm
Crack	
Count	1
Dimensions	$1 \times 25 \times 1$ mm
Centre	$(0, 0)$ mm
Notch root radius	1 mm
PZT discs	
Count	2
Dimensions	$r = 6.35$ mm, $t = 0.254$ mm
Actuator position	$(-69.85, 0, -1.6)$ mm
Sensor position	$(+69.85, 0, +1.6)$ mm

Table 5: Material data for the plate and the piezoelectric discs.

Property	Aluminium plate	PZT disc
Material	Al 2024-T3	APC850
Density	2700 kg/m^3	7600 kg/m^3
Young’s modulus	69 GPa	63 GPa
Poisson’s ratio	0.33	0.35
Permittivity	—	$1.6851 \times 10^{-8} \text{ F/m}$
Charge constants	—	$d_{15} = d_{24} = 590 \text{ pC/N}$
	—	$d_{31} = d_{32} = -175 \text{ pC/N}$
	—	$d_{33} = 400 \text{ pC/N}$

All remaining components of the charge-constant matrix are zero.

The distance between the two discs is 139.7 mm. With the S_0 group velocity of Table 3, the direct arrival is expected at $139.7/5332 = 26.2 \mu\text{s}$, which is worth remembering when the signals are examined in Section 4.5.

4.2. Partitioning and mesh

Meshing a plate with two circular patches bonded to it is the first practical obstacle. A free mesh degrades badly where it has to transition around a

circular footprint, and the elements it produces there are distorted enough to reflect the wave on their own. The plate was therefore partitioned before meshing, both around the transducer footprints and around the crack, so the mesh could be built from mappable regions (Figures 1 and 2). Edge seeding was then used to refine the mesh locally near the discs and near the crack, where the wavelength of the scattered field is shortest.

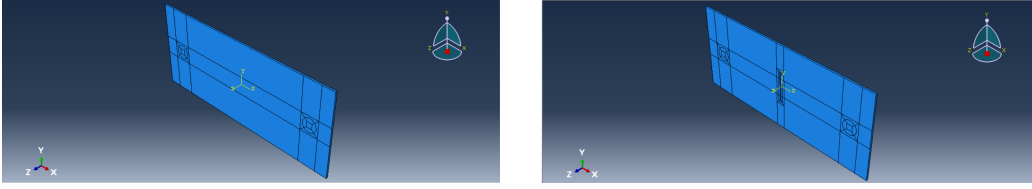


Figure 1: Plate partitioning before meshing. Left: healthy plate, with the two circular transducer footprints partitioned into mappable faces. Right: the same plate with the 25 mm crack partition added at the centre.

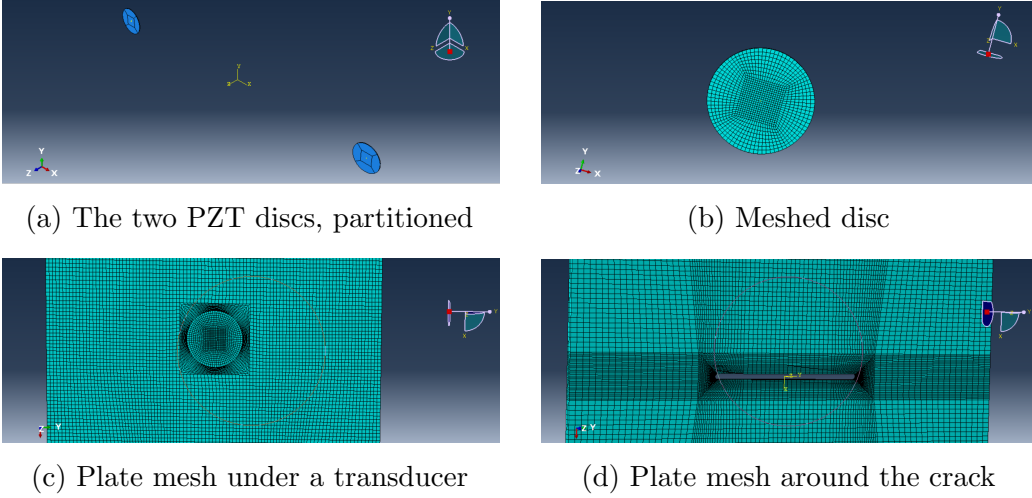


Figure 2: Discretisation of the two-transducer model. The mesh is refined locally where the geometry changes and where the scattered field is expected to have the shortest wavelength.

4.3. Loads, boundary conditions and solution strategy

No mechanical load is applied to the plate or to the discs. The excitation is entirely electrical. The bonded face of each disc, where it meets the plate,

is held at zero volts, and the free face of the actuator is driven with a 200 kHz sinusoidal potential of 10 V amplitude. The free face of the sensor is left free to develop a potential, and that potential is the output of the analysis.

The solver strategy is dictated by an Abaqus restriction. The fully coupled piezoelectric continuum element is available in Abaqus/Standard but not in Abaqus/Explicit, whereas explicit time integration is what makes a wave-propagation analysis affordable. The two requirements were reconciled by splitting the problem: the aluminium plate is solved with Dynamic, Explicit, the PZT discs with Dynamic, Implicit, and the two analyses are run simultaneously and exchange data at their common interface through an Abaqus co-execution (Figure 3). Getting the interface right, so that the partitioned faces on the underside of each disc and on the surface of the plate coincide, is the part of the setup that most often prevents the analysis from running.

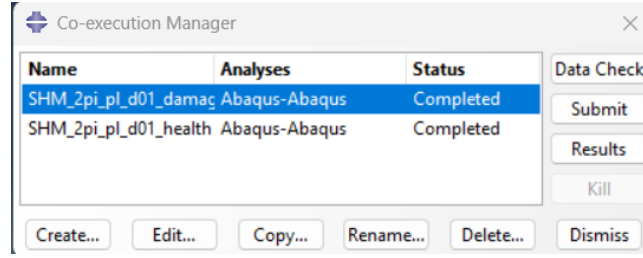


Figure 3: Co-execution manager. The healthy and cracked cases are each solved as a coupled Abaqus/Standard–Abaqus/Explicit pair, both completing successfully.

Two analyses were run, identical except for the crack: a healthy plate and a plate in which the crack was introduced as a cut extruded through the thickness. Both were run to 80 μ s.

4.4. Damage indices considered

Once a healthy signal $S_r(i)$ and a current signal $S_d(i)$ are available, a scalar damage index has to be extracted from them. Several were reviewed before two were selected.

Signal difference coefficient (SDC).. Defined as one minus the normalised cross-correlation of the two signals,

$$\text{SDC} = 1 - \frac{\sum_{i=1}^N \tilde{S}_d(i) \tilde{S}_r(i)}{\sqrt{\sum_{i=1}^N \tilde{S}_d^2(i) \sum_{i=1}^N \tilde{S}_r^2(i)}} \quad (6)$$

where $\tilde{S}(i) = S(i) - \bar{S}$ is the mean-removed signal and the sums run over the N samples of the record. Identical signals give $\text{SDC} = 0$ and completely dissimilar signals give $\text{SDC} = 1$. Because it uses the full waveform, SDC responds to changes in shape and phase as well as amplitude.

Normalised root-mean-square deviation..

$$\text{NRMSD} = \sqrt{\frac{\sum_{i=1}^N [S_d(i) - S_r(i)]^2}{\sum_{i=1}^N S_r(i)^2}} \quad (7)$$

which measures the energy of the difference between the signals, normalised by the energy of the baseline. It has a clear physical reading as a fraction of signal energy changed, and thresholds around 0.2 are commonly quoted as the level above which a change exceeds normal environmental variation.

Peak amplitude ratio.. The simplest index of all,

$$A_{\text{ratio}} = \frac{A_{\text{damaged}}}{A_{\text{baseline}}} \quad (8)$$

comparing the peak amplitude of the first arriving packet. Scattering at the crack removes energy from the direct path, so a ratio below one indicates attenuation.

Other features.. Time-of-flight shift, residual signal energy, and time-frequency features from the continuous wavelet transform, the short-time Fourier transform or the Hilbert envelope were also reviewed. The wavelet-energy feature is the one that becomes central in Stage 2, and it is defined in Section 5.7.

4.5. Results

Figure 4 shows the wave leaving the actuator and spreading across the plate, confirming that the electrical excitation does generate a propagating elastic field. Figure 5 shows the electric potential recorded at the sensor electrode in each of the two states.

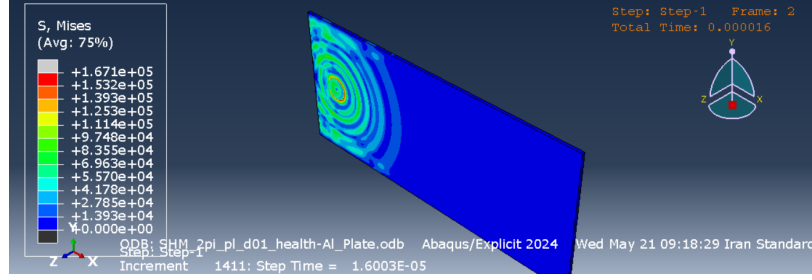
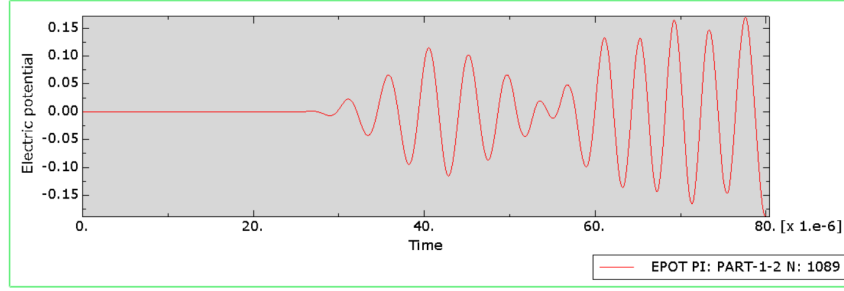
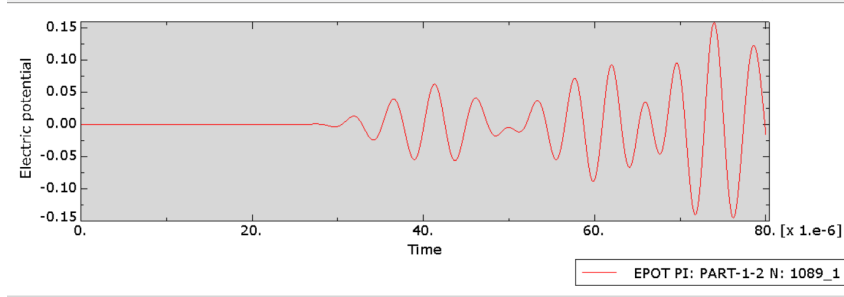


Figure 4: Von Mises stress at $16 \mu\text{s}$, showing the wave radiating from the actuator across the healthy plate.



(a) Healthy plate



(b) Cracked plate

Figure 5: Electric potential at the sensor electrode over $80 \mu\text{s}$. Both traces are flat until about $27 \mu\text{s}$, which agrees with the $26.2 \mu\text{s}$ predicted from the S_0 group velocity.

Both signals stay at zero until roughly $27 \mu\text{s}$ and then begin to oscillate. That arrival time is the first useful check on the model: the S_0 group velocity from Table 3 predicts $26.2 \mu\text{s}$ over the 139.7 mm path, so the wave in the

model is travelling at the speed the analytical dispersion relation says it should.

Overlaying the two traces (Figure 6) shows the effect of the crack. The cracked-state signal is lower in amplitude throughout, and the two waveforms drift progressively out of phase as time increases.

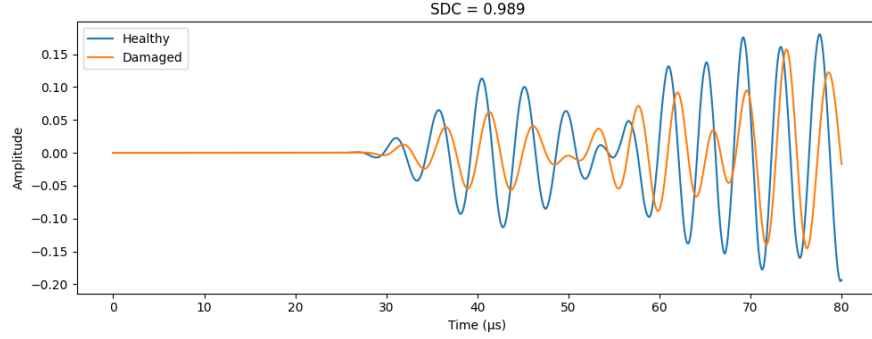


Figure 6: Sensor signal from the healthy plate and from the cracked plate, overlaid. The crack reduces the amplitude and shifts the phase of the later part of the record.

The two selected indices give the values in Table 6.

Table 6: Damage indices for the two-transducer model.

Quantity	Value
Signal difference coefficient	0.989
Peak amplitude, healthy	0.195
Peak amplitude, cracked	0.157
Amplitude drop	19.5%

The amplitude drop of 19.5% is unambiguous. A change of 5–10% is normally taken as suggestive of damage and anything above 10% as a strong indication, so a drop of nearly 20% is a clear detection. The SDC of 0.989 is also a clear detection, but its magnitude needs care in interpretation. Equation (6) is evaluated at zero lag over the whole 80 μs record, and because the two waveforms drift out of phase towards the end of that window their zero-lag correlation collapses towards zero, which sends SDC towards one. The index has effectively saturated. It confirms that the signals differ, but at

that value it can no longer rank one difference against another, which matters as soon as several sensing paths have to be compared.

4.6. From detection to localisation

The two-transducer model was built to answer one question, and by design it carries no information for a second one. Three specific requirements separate what it does from what localisation needs, and Stage 2 is built around meeting all three.

The first is geometric. A single actuator-sensor pair yields one number, and one number cannot locate anything. The crack in this model happens to sit on the direct path, so it produces a large index; a crack of the same size sitting 30 mm to the side would produce a smaller index, and the two situations are indistinguishable from a single measurement. A single path can therefore only detect, and reaching Level 2 of Table 1 requires several paths crossing the monitored area at different angles.

The second is in the excitation. Driving the actuator with a continuous 200 kHz sinusoid, as was done here, means the plate is being fed energy for the entire analysis. The received signal grows steadily throughout the record rather than arriving as a discrete packet, and by 80 μ s it also contains reflections from the plate edges superposed on the direct arrival. That is workable when the only question is whether the signal changed, and it is unusable when the arrival time of a specific wave packet has to be measured. A windowed tone burst is required instead, and that is what Stage 2 uses.

The third is a modelling issue. The crack here was created by cutting material out of the plate, which means the healthy and cracked models have different meshes. Part of the difference between the two signals is therefore attributable to the change in discretisation rather than to the crack. For a detection threshold this is tolerable, and for a quantitative per-path index that has to be compared across six paths it is not.

5. Stage 2: crack localisation with a four-transducer network

The Stage 2 model extends the same finite-element approach to meet all three requirements above.

5.1. Layout and geometry

Four PZT discs are placed at the corners of a square on a $400 \times 400 \times 1.6$ mm aluminium plate. Four transducers give $\binom{4}{2} = 6$ unique sensing paths, four

along the sides and two along the diagonals. The crack is placed off-centre, on the P1–P3 diagonal, so that one path passes directly through it while the remaining five pass at increasing perpendicular offsets. Table 7 gives the layout and Figure 7 shows it.

Table 7: Geometry of the four-transducer model.

Plate	$400 \times 400 \times 1.6$ mm, aluminium
PZT discs	4, diameter 12.7 mm, thickness 0.254 mm
PZT positions	P1(−75, −75), P2(75, −75), P3(75, 75), P4(−75, 75) mm
Crack	25 mm long, through-thickness, centred at (−35, −35) mm
Path lengths	4 sides at 150 mm, 2 diagonals at 212.1 mm

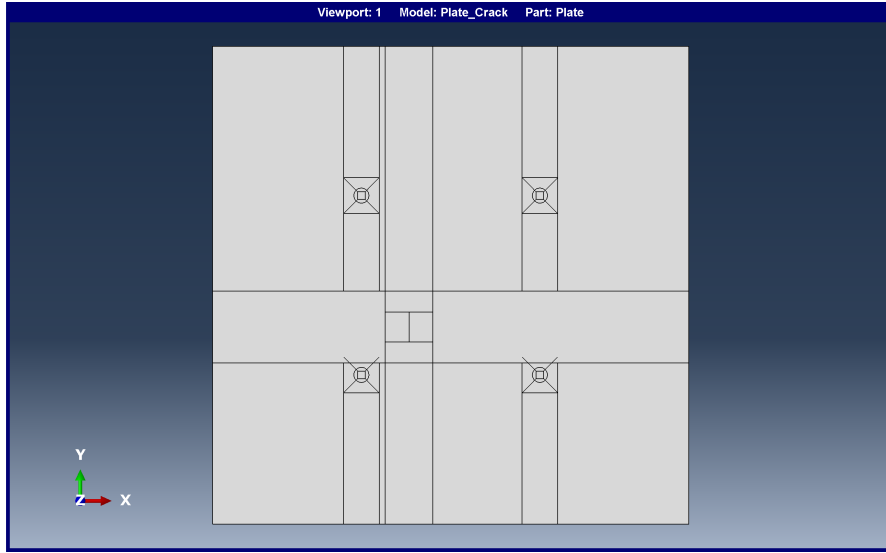


Figure 7: Plate part seen from above. Each of the four transducer footprints carries the butterfly partition inside its 30×30 mm structured box, and the small partition left of centre bounds the crack.

The plate size follows from a timing constraint. This is the smallest plate that keeps the first edge reflection outside the S_0 arrival window on all six paths: reflections arrive at $54.7 \mu\text{s}$, while the latest S_0 window closes at $48.1 \mu\text{s}$. That removes the edge-reflection contamination which limited the Stage 1 record.

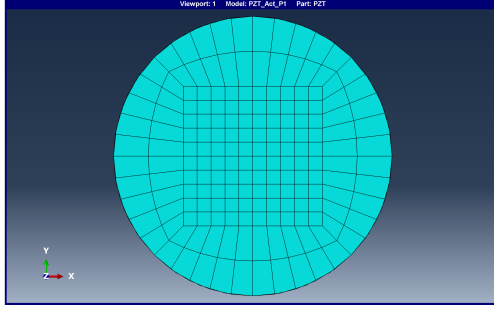
5.2. Crack representation

Instead of cutting material away, the crack is modelled as a zero-thickness seam. A seam duplicates the nodes on the crack face so the two faces can separate, while leaving the rest of the mesh untouched. The healthy and cracked models therefore have identical meshes, element for element: both contain 1,040,892 elements, and the seam adds exactly 150 duplicated nodes on the crack face. The only difference between the two analyses is the crack itself, which is what a per-path damage index requires if it is to mean anything. A notch variant with a 1 mm wide cut was also built in the same CAE model, giving a second crack representation to compare the seam against; its co-simulations were not run for this report.

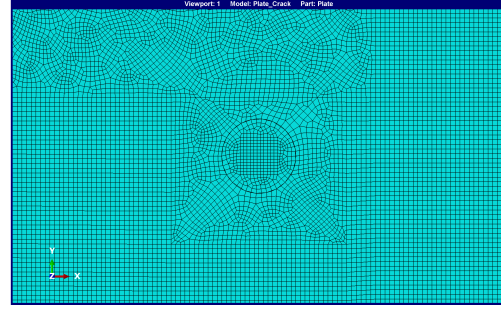
5.3. Mesh design from the dispersion relations

The element size comes straight out of Table 3. The shorter of the two wavelengths at 200 kHz belongs to A_0 at 7.90 mm, and the usual requirement is ten elements per wavelength, giving an in-plane element size of 0.8 mm, with four elements through the thickness.

Meshing the plate with four circular footprints raised the same difficulty as in Stage 1, in a more severe form. A free mesh produced 224 elements with zero or negative volume and 952 elements with an aspect ratio above 100, and it also introduced a spurious reflection at the mesh transition that was visible in the signals. The fix was to partition each footprint into a butterfly pattern of five mappable faces, an inner square plus four radial segments reaching the corners of a 30×30 mm structured box, and to apply the same partition to the discs themselves (Figure 8). The plate away from the boxes was then meshed with a structured technique and only the boxes with sweep or advancing-front. The result is a fully structured hexahedral mesh of C3D8E elements with zero failed and zero warning elements, and the spurious reflection disappeared with it.



(a) Meshed PZT disc



(b) Plate mesh beneath transducer P1

Figure 8: Butterfly partition into an inner square and four radial segments, applied to the disc itself (a) and to the plate under it (b). The inner square meshes as a regular grid, and the four segments carry the transition out to the edge of the 30×30 mm structured box. That box edge is the visible boundary in (b) between the swept region and the uniform 0.8 mm grid outside it.

A seam has zero thickness, so the mesh closes over it and the crack leaves nothing to see from above. Figure 9(a) therefore marks the seam on the surrounding mesh, while Figure 9(b) shows the notch variant, where the 1 mm cut is a real hole in the mesh.

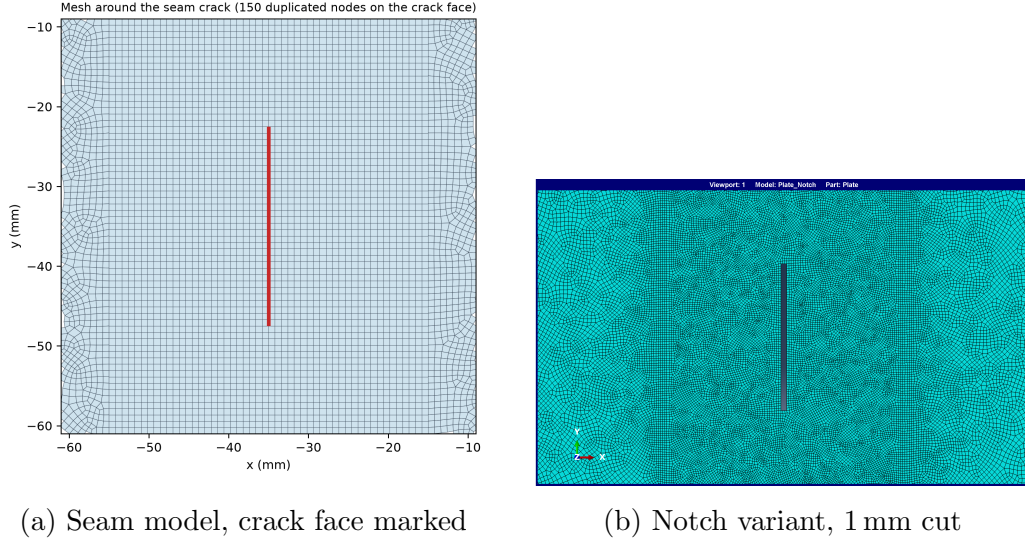


Figure 9: The crack region. In (a) the uniform 0.8 mm mesh runs through the crack location undisturbed, with the seam drawn in red along the line where its faces are duplicated. In (b) the notch variant removes material, so the mesh wraps around the cut and the element count rises to 1,231,288.

5.4. Excitation

The continuous sinusoid of Stage 1 is replaced by a four-cycle Hann-windowed tone burst,

$$V(t) = \frac{V_0}{2} \left[1 - \cos\left(\frac{2\pi f_0 t}{N_{\text{cyc}}}\right) \right] \sin(2\pi f_0 t), \quad 0 \leq t \leq \frac{N_{\text{cyc}}}{f_0} \quad (9)$$

with $V_0 = 10 \text{ V}$, $f_0 = 200 \text{ kHz}$ and $N_{\text{cyc}} = 4$. The window narrows the bandwidth, which limits dispersion of the packet, and it produces a wave packet with a definite start and end, which is what makes an arrival time measurable.

5.5. Electrode modelling and co-simulation

The bottom electrode of every disc is grounded. The actuator's top electrode is driven with the tone burst. For the sensors, the top electrode is tied to a single equipotential node through `*EQUATION` constraints on the electric-potential degree of freedom, which reproduces the behaviour of a real continuous metal electrode instead of treating every finite-element node

on the surface as an independent voltage output (Figure 10). The electric potential at each electrode's master node is what is recorded.



Figure 10: Electrical boundary conditions on one transducer: grounded bottom electrode, driven or sensing top electrode.

The plate is again solved in Abaqus/Explicit and the discs in Abaqus/Standard, coupled through a surface-based co-simulation with lockstep incrementation. One co-simulation is required per actuator per plate state, so eight runs were performed in total, covering actuators P1 to P4 in both healthy and cracked states. Three actuators are enough to cover all six unique undirected paths; the fourth actuator repeats those paths in the reverse direction and was used as an independent reciprocity check.

5.6. Verification

The following checks were performed before any result was interpreted.

- Healthy and cracked meshes are identical element for element, 1,040,892 elements each, with the seam contributing exactly 150 duplicated nodes.
- Mesh quality verification with analysis checks reports zero failed and zero warning elements on all plate variants.
- A data check on the plate model and on each disc model completes with zero errors. This caught two real modelling errors during development, a duplicate equation-constraint conflict created by the butterfly partition

and a missing local material orientation on the anisotropic piezoelectric material, both fixed before the production runs.

- All eight co-simulations reached completed status.

5.7. Damage index and probability-based diagnostic imaging

Following Nasiri et al. [25], the damage-sensitive feature is the wavelet energy of the signal inside a time window,

$$F = \sum_t \sum_f |W(t, f)|^2 \quad (10)$$

where $W(t, f)$ is the continuous wavelet transform of the signal using a complex Morlet wavelet. The damage index for a path is the relative change in that energy,

$$\text{DI} = \frac{|F_{\text{damaged}} - F_{\text{healthy}}|}{F_{\text{healthy}}}. \quad (11)$$

A wavelet-energy feature was preferred over the SDC used in Stage 1 for the reason given in Section 4.5: SDC saturates, and a per-path index that has to be compared across six paths must stay graded.

Turning six scalar indices into a two-dimensional map is the job of the PDI algorithm. Each path contributes an elliptical weight function centred on the straight line joining its two transducers,

$$r(x, y) = \frac{d_i + d_j}{d_{ij}} - 1 \quad (12)$$

$$\gamma = \frac{V_{S_0} T_{S_0}}{d_{ij}} \quad (13)$$

$$\beta = \frac{\sqrt{2}}{n} \gamma, \quad n = 4 \quad (14)$$

$$W(x, y) = \exp \left[- \left(\frac{r}{\beta} \right)^2 \right] \quad (15)$$

$$P(x, y) = \sum_{\text{paths}} \text{DI}_{ij} W_{ij}(x, y) \quad (16)$$

where d_i and d_j are the distances from the point (x, y) to the two transducers, d_{ij} is the direct path length, and $T_{S_0} = N_{\text{cyc}}/f_0$ is the duration of the S_0 packet. The quantity r is zero on the direct path and grows away from it, so each path produces a band of high weight that is widest at the midpoint and narrows towards the transducers. Summing the weighted indices over all paths concentrates probability where the bands from the most strongly affected paths overlap. The coefficient β controls how wide those bands are. Tying β to the physical length of the wave packet, which is what [25] contributes, replaces the trial-and-error value used in earlier formulations.

5.8. Results

5.8.1. Sensor signals

Figure 11 compares the received signal on all six paths, healthy against cracked, ordered by how close each path passes to the crack. The vertical marker on each subplot is the S_0 arrival time predicted from the group velocity of Table 3, and it lands on the observed onset in every case.

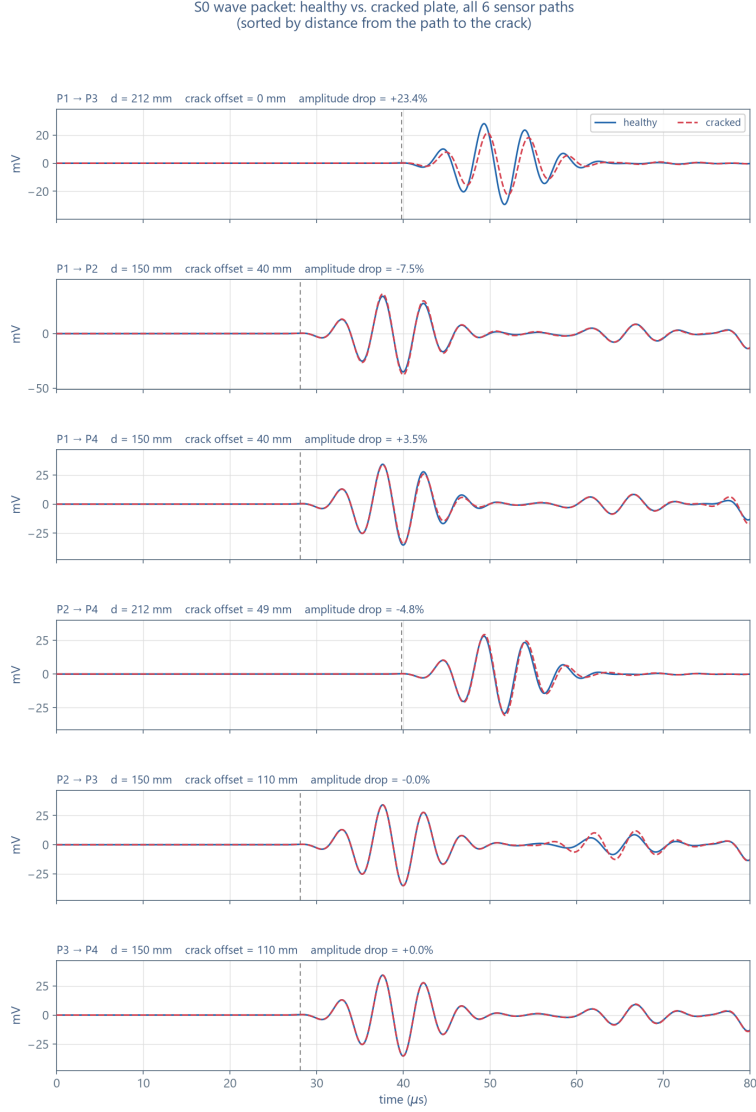


Figure 11: S_0 wave packet on all six sensing paths, healthy plate against cracked plate. The predicted S_0 arrival time is marked on each subplot.

The pattern across the six paths follows the crack geometry (Figure 12 and Table 8). The P1–P3 diagonal, which passes straight through the crack at zero offset, loses 23.4% of its peak amplitude and changes shape after the first arrival. The three paths offset by 40–50 mm change by a few percent, in either direction, which is consistent with the scattered field interfering

constructively or destructively at the sensor depending on the path geometry. The two paths 110 mm from the crack show no change at all, to four decimal places in the damage index. That last observation also confirms the healthy and cracked models are identical apart from the crack, since meshes that differed would have registered something on those distant paths.

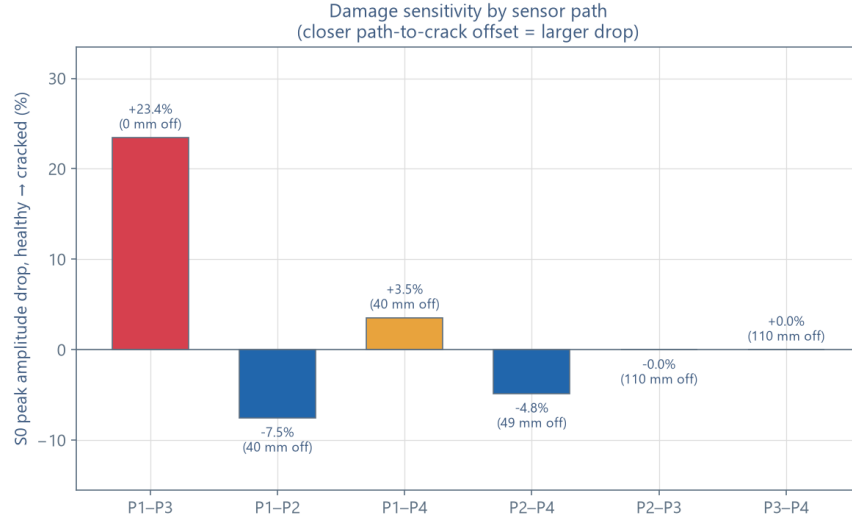


Figure 12: Peak amplitude drop from healthy to cracked state for each sensing path, annotated with the path’s perpendicular offset from the crack.

Table 8: Per-path damage indices and amplitude drops. A negative drop means the amplitude increased.

Path	Length (mm)	Offset from crack (mm)	Damage index	Amplitude drop (%)
P1-P3	212.1	0	0.4065	23.4
P1-P2	150.0	40	0.1400	−7.5
P1-P4	150.0	40	0.0696	3.5
P2-P4	212.1	49.5	0.0996	−4.8
P2-P3	150.0	110	0.0001	0.0
P3-P4	150.0	110	0.0002	0.0

5.8.2. Localisation by probability-based diagnostic imaging

Feeding the six damage indices into the PDI algorithm produces the map in Figure 13. The peak of the map is at $(-35, -45)$ mm against a true crack

location of $(-35, -35)$ mm, an error of 10.0 mm. On a plate spanning 400 mm that is an error of about 2.5%, obtained from four transducers and six paths.

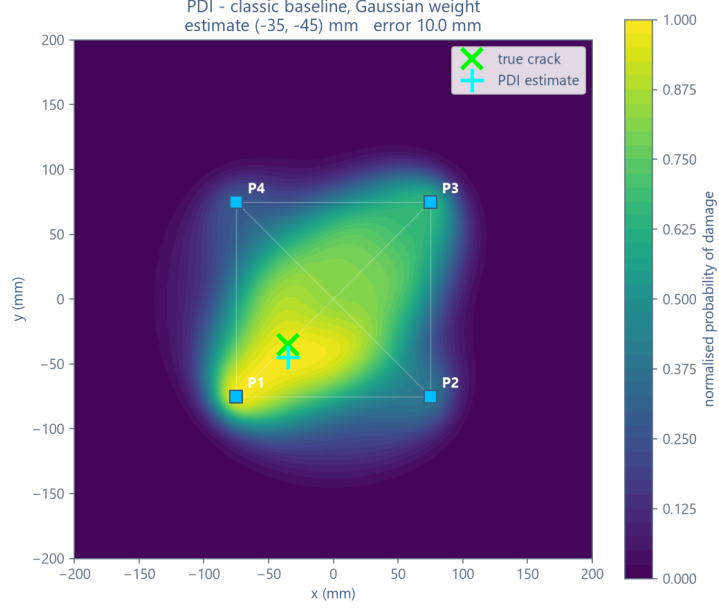


Figure 13: Damage-probability map from the PDI algorithm with Gaussian elliptical weighting. Green cross: true crack location. Cyan cross: the peak of the map, which is the algorithm’s estimate.

5.8.3. Localisation by time-of-flight delay-and-sum

A second and independent estimator was implemented on the same extracted signals, with no further simulation. Subtracting the healthy signal from the cracked signal on each path,

$$v_{\text{residual}}(t) = v_{\text{cracked}}(t) - v_{\text{healthy}}(t), \quad (17)$$

removes the direct arrival, which is common to both states, and leaves the field scattered by the crack. For every candidate point on the plate the expected time of flight from actuator to that point to sensor is computed, the residual envelope is sampled at that delay, and the contributions from all six paths are summed. This is a migration image, the same principle as the total focusing method used in phased-array ultrasonic inspection.

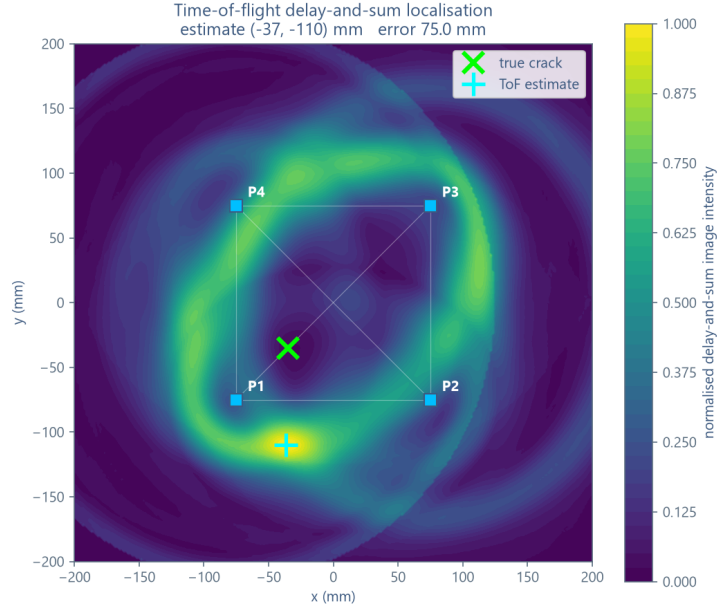


Figure 14: Delay-and-sum image formed from the time of flight of the crack-scattered wave on all six paths.

The estimate is $(-37, -110)$ mm, an error of 75.0 mm, which is much worse than the PDI result. Two effects account for this. The four-cycle tone burst lasts about $20\text{ }\mu\text{s}$, which at the S_0 group velocity corresponds to a wave packet roughly 100 mm long in space, so the envelope peak used to measure the arrival is blurred over tens of millimetres whatever method reads it. On top of that, four transducers at the corners give six paths with limited angular diversity, and the image cannot distinguish the true crossing point from other places where several path ellipses happen to intersect. The bright ring away from the crack in Figure 14 is that ghosting, a known behaviour of sparse-array delay-and-sum imaging. The full-field check below establishes that the underlying model and signals are sound.

5.8.4. Full-field validation

The last check verifies the model itself. It is not deployable, since it requires the entire simulated displacement field, which no realistic sensor network can measure. The whole-plate displacement field was extracted from the explicit output database at $t = 40\text{ }\mu\text{s}$ for both the healthy and cracked models under actuator P1, and the two fields were subtracted node by node.

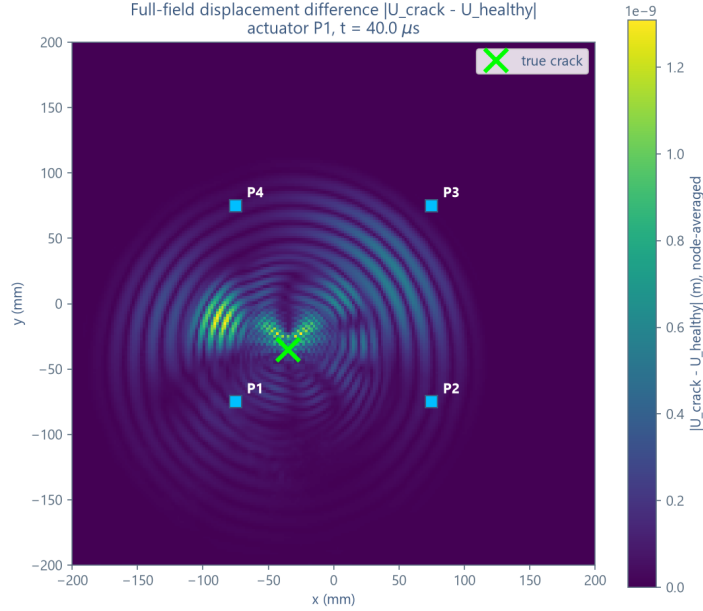


Figure 15: $|U_{\text{crack}} - U_{\text{healthy}}|$ over the whole plate at $t = 40 \mu\text{s}$. The circular wavefront is centred on the true crack location.

The difference field is a clean circular wavefront expanding from a point that coincides with the true crack location to within the 2 mm resolution of the grid used. The crack behaves as a secondary point source exactly where it was placed. What the point-sensor methods infer indirectly from six voltage traces is therefore visible directly here, and it puts an upper bound on how much of the 10 mm PDI error the model itself could account for: the model places the crack correctly, so the remaining error is set by what four sensors and six paths can resolve.

5.9. Comparison of the three estimators

Table 9: Localisation error of the three methods applied to the same simulation.

Method	Input used	Localisation error
Classical PDI (wavelet energy)	6 paths + healthy baseline	10.0 mm
Time-of-flight delay-and-sum	6 paths + healthy baseline	75.0 mm
Full-field wavefield difference	full displacement field	~ 2 mm (validation only)

Three points follow from Table 9.

The energy change in a guided-wave signal correlates strongly with how close the direct path passes to the crack. The path that intersects the crack changes by far the most, and paths 110 mm away do not change at all. Elliptical weighting turns that correlation into a usable image, and it is the best-performing sensor-based method tested here.

Changing the feature from amplitude to time of flight did not improve accuracy with this layout. With four corner transducers the aperture is too sparse to separate the true crossing point from ghost intersections, and the finite duration of the tone burst blurs any envelope-based time measurement regardless of which feature is used. The elliptical Gaussian weight in the PDI algorithm already absorbs that blur, which is why it tolerates the sparse layout better.

The full-field check settles where the residual error comes from. The crack sits where it was placed in the model, so the 10 mm is what this sensor layout can resolve.

5.10. Comparison against published PDI results

The PDI algorithm used here is the one developed by Nasiri et al. [25] and applied to sensor-layout design by Keshavarz Motamed et al. [26], and both report actual localisation errors that this result can be checked against.

Nasiri et al. instrument a $500 \times 500 \times 1$ mm stiffener-reinforced aluminium plate with nine PZTs (56 sensing paths) and report errors of 2.2–41.9 mm across three single-crack scenarios with their best weight function (β -GWD), averaging about 22 mm; their fatigue cracks are 4 mm long. Keshavarz Motamed et al. use the same $400 \times 400 \times 1$ mm aluminium plate geometry as this report, with a 200×200 mm central monitoring area and a 20 mm crack, and report two relevant results: with 8 sensors, finite-element verification of three single-crack cases gives errors of 7.8–19.5 mm; and, across a statistical sweep of 121 crack positions, a circular array needs 14–20 sensors to keep 80% of predictions within 10 mm.

Table 10: Published PDI localisation errors compared against the result of Section 5.

Source	Plate	Crack	Sensors	Localisation error
Nasiri et al. [25]	500×500 mm, stiffened	4 mm	9 (56 paths)	2.2–41.9 mm (3 cases)
Motamed et al. [26]	400×400 mm, plain	20 mm	8	7.8–19.5 mm (3 cases)
Motamed et al. [26]	400×400 mm, plain	20 mm	14–20	10 mm at 80th pctl.
This report (Section 5)	400×400 mm, plain	25 mm	4 (6 paths)	10.0 mm (1 case)

The comparison has to be read with its caveats attached, and the crack itself is the largest one: at 25 mm, the crack in this report is roughly six times longer than the 4 mm fatigue cracks in Nasiri et al. and about a quarter longer than the 20 mm crack in Keshavarz Motamed et al. A longer crack scatters more of the wave and is intrinsically easier to detect and locate, so part of the gap in sensor count is the comparison rewarding a larger flaw rather than a better method. The crack position adds a second advantage: it sits exactly on the P1–P3 path (Table 8), which is the single most favourable geometry for any of these arrays, whereas the published numbers are statistics over many crack positions and orientations, most of which are not that well aligned. Nasiri et al.’s plate also carries a stiffener and two rows of rivet holes, which give the wave field more to scatter from than the plain plate used here. Taken together, this report’s 10.0 mm is one data point from a favourable case, not a distribution, and it should be read as evidence that the PDI method can reach this accuracy with four sensors under good conditions, not as evidence that four sensors are sufficient for this plate in general. Within that reading, the result is still a genuine one: a bachelor’s-level model with a sparse, unoptimised four-corner layout reaches the same accuracy that the cited graduate-level studies, from the same research group, needed considerably denser arrays to reach even on their easiest cases.

6. Discussion

Read together, the three phases form a single argument: choose the method, demonstrate that it detects, then extend it until it localises.

The method review fixed the direction of the work. Once the requirement was stated as high sensitivity to a small crack, a large monitored area per sensor, and permanent installation, guided-wave monitoring with bonded piezoelectric transducers was the only family that satisfied all three. The review supplies the argument for that choice.

The two-transducer model established that the physics could be captured numerically. The arrival time matched the analytical S_0 group velocity, the crack produced a 19.5% amplitude drop, and the implicit-explicit co-execution proved to be a workable route around the absence of coupled piezoelectric elements in Abaqus/Explicit. It also made clear three specific requirements a localisation-capable model would have to satisfy: multiple paths, since a single path cannot localise; a windowed excitation, since a continuous tone prevents any time-of-flight measurement and admits edge reflections; and a crack representation that leaves the mesh unchanged, since a crack cut out of the mesh contaminates the comparison with a discretisation difference.

The four-transducer model meets all three. Six paths replaced one, a windowed tone burst replaced the continuous sinusoid, and a seam replaced the cut so that the healthy and cracked meshes stayed identical. That last change buys a piece of verification the Stage 1 model could not offer: with the two meshes identical element for element, the damage index of 0.0001 reported by the two distant paths measures the numerical noise floor of the whole procedure. Against that floor, the 0.4065 index on the path through the crack is unambiguous.

The gap between the PDI result and the delay-and-sum result is what the comparison teaches. Both estimators consume the same six signals, and one is seven times more accurate than the other, so the difference lies in how much each method demands of the data. Delay-and-sum needs a resolvable arrival time from every path, which a 100 mm-long wave packet on a sparse array cannot supply. PDI needs only a graded scalar per path, which the same signals do supply. Array density, then, is what limits accuracy here, and switching damage feature will not move it much.

7. Conclusions

1. A comparison of SHM method families against the requirements of a thin metallic panel selects guided-wave monitoring with bonded piezoelectric transducers, which is the only approach combining sensitivity to small cracks with a large monitored area per sensor and permanent installation.
2. A coupled finite-element model of an aluminium plate with two bonded PZT discs, solved as an implicit-explicit co-execution in Abaqus, reproduces S_0 Lamb-wave propagation at the velocity predicted by the Rayleigh–Lamb equations. A 25 mm crack on the direct path produces a 19.5% drop in

peak amplitude and a signal difference coefficient of 0.989, which is a clear Level 1 detection.

3. The same modelling approach extended to four transducers gives six independent sensing paths. Damage indices computed from the continuous wavelet energy of each path follow the expected physical ordering, from 0.4065 on the path through the crack down to 0.0001 on the paths 110 mm away.
4. Combining those indices through a probability-based diagnostic imaging algorithm with elliptical Gaussian weighting locates the crack to within 10 mm of its true position on a 400 mm plate, which is a Level 2 diagnosis from four sensors. That error sits inside the 2.2–41.9 mm range reported for the same class of algorithm with denser arrays in two published studies (Section 5.10), one of which uses the identical 400×400 mm plate and needs 14–20 sensors for a comparable error at its 80th percentile, against the four used here on a single, favourably positioned case with a longer crack.
5. A time-of-flight delay-and-sum image built from the same signals gives a 75 mm error, caused by the sparse aperture of a four-corner array together with the spatial length of the tone burst.
6. A full-field wavefield-difference check shows the crack acting as a secondary point source at its true location to within the 2 mm grid resolution. The residual localisation error is therefore set by what the sensor layout can resolve.

8. Future work

Denser and better-distributed arrays.. Table 9 points to array sparsity as the factor limiting accuracy. Six, eight or nine transducers distributed around the plate, instead of four at the corners, would increase both the number of independent paths and the angular diversity. The delay-and-sum estimator, whose ghosting is a direct consequence of too few paths, would gain the most.

Feature fusion.. Amplitude-based and time-of-flight features fail in different ways, the first through interference of the scattered field and the second through sparse-aperture ghosting. Combining them into a single fused image would prevent either failure mode from dominating.

Baseline-free operation.. Every index used here needs a recorded healthy signal, and in service that baseline drifts with temperature and loading. Instantaneous-baseline methods, which compare geometrically identical paths within the same measurement instead of comparing against a stored record, remove that dependence [22, 23, 24], and they are the natural next step for making this approach usable outside a simulation.

Optimal sensor placement.. The layout can be treated as a design variable and optimised against localisation accuracy [26, 16], which turns the trial layout used here into a designed one.

Machine learning on the signal database.. The finite-element model can generate labelled signals for many crack positions, lengths and orientations at a cost that physical experiments cannot match. That database is what a learned classifier or regressor needs, and it would let damage type and severity, Levels 3 and 5 of Table 1, be estimated alongside position [17, 8].

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